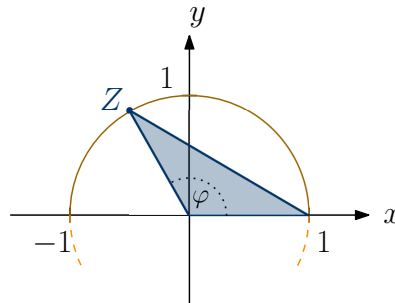


Solutions to Problem Set 7

1. Expectation of transformed random variables

- (a) 1 \$ is put on the table. We toss a coin that shows “heads” with probability $p \in (0, 1)$. Every time the coin shows “heads”, the amount of money on the table is doubled, and we stop the game after n steps. Let X be the amount of money on the table after the experiment is over. Calculate $\mathbf{E}[X]$ and $\text{Var}[X]$.
- (b) Consider a circle centered at $(0, 0)$ with radius 1. A point Z is chosen uniformly on the upper half of the circle. Let A denote the area of the triangle with coordinates $(0, 0)$, $(1, 0)$ and Z (see the sketch below). Calculate $\mathbf{E}[A]$ and $\text{Var}[A]$.



Hint: Represent Z in polar coordinates, $Z = (\cos(\Phi), \sin(\Phi))$, where $\Phi \sim \mathcal{U}([0, \pi])$. Use the identity $\cos(2\varphi) = 1 - 2\sin^2(\varphi)$ for $\varphi \in \mathbb{R}$.

- (a) Let $Y \sim \text{Bin}(n, p)$. The amount of money on the table at the end is $X = 2^Y$. We want to calculate $\mathbf{E}[X]$ and $\text{Var}[X]$. We use the probability mass function of a binomial distribution

$$p_Y(k) = \mathbf{P}[Y = k] = \binom{n}{k} p^k (1-p)^{n-k}.$$

We thus find that

$$\begin{aligned} \mathbf{E}[X] &= \mathbf{E}[2^Y] = \sum_{k=0}^n 2^k p_Y(k) = \sum_{k=0}^n \binom{n}{k} (2p)^k (1-p)^{n-k} \\ &= ((1-p) + 2p)^n = (1+p)^n. \end{aligned}$$

For the calculation of $\text{Var}[X]$, we first calculate $\mathbf{E}[X^2]$:

$$\begin{aligned} \mathbf{E}[X^2] &= \mathbf{E}[4^Y] = \sum_{k=0}^n 4^k p_Y(k) = \sum_{k=0}^n \binom{n}{k} (4p)^k (1-p)^{n-k} \\ &= ((1-p) + 4p)^n = (1+3p)^n. \end{aligned}$$

Therefore we have

$$\text{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = (1+3p)^n - (1+p)^{2n}.$$

- (b) Representing Z in polar coordinates, we have $Z = (\cos(\Phi), \sin(\Phi))$, and $\Phi \sim \mathcal{U}([0, \pi])$. The law of Φ therefore has the probability density function

$$f_{\Phi}(\varphi) = \frac{1}{\pi} \mathbb{1}_{[0, \pi]}(\varphi).$$

The area of the triangle is thus given by $A = g(\Phi) = \frac{1}{2} \sin(\Phi)$ (with $g(\varphi) = \frac{1}{2} \sin(\varphi)$). We can immediately calculate

$$\begin{aligned} \mathbf{E}[A] &= \mathbf{E}[g(\Phi)] = \int_{-\infty}^{\infty} g(\varphi) f_{\Phi}(\varphi) d\varphi \\ &= \frac{1}{\pi} \int_0^{\pi} \frac{1}{2} \sin(\varphi) d\varphi = \frac{1}{\pi} [-\cos(\varphi)]_0^{\pi} \\ &= \frac{1}{2\pi} (\cos(0) - \cos(\pi)) = \frac{1}{\pi}. \end{aligned}$$

For the variance:

$$\begin{aligned} \mathbf{E}[A^2] &= \frac{1}{\pi} \int_0^{\pi} \frac{1}{4} \sin^2(\varphi) d\varphi = \frac{1}{\pi} \int_0^{\pi} \left(\frac{1 - \cos(2\varphi)}{2} \right) d\varphi \\ &= \frac{1}{8} + \frac{1}{8\pi} [-\sin(2\varphi)]_0^{\pi} = \frac{1}{8}. \end{aligned}$$

We thus see that

$$\text{Var}[A] = \mathbf{E}[A^2] - \mathbf{E}[A]^2 = \frac{1}{8} - \frac{1}{\pi^2}.$$

2. Markov and Čebyšev inequalities

- (a) A biased coin lands on heads with probability $\frac{1}{3}$. This coin is tossed 200 times. Use Markov's inequality to give an upper bound on the probability that the coin shows heads at least 120 times. Improve this bound using Čebyšev's inequality.
- (b) Find a random variable $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and a value $a > 0$ such that Čebyšev's inequality is tight, i.e.:

$$\mathbf{P}[|X - \mathbf{E}[X]| \geq a] = \frac{\text{Var}[X]}{a^2}.$$

Hint: It suffices to determine the law $\mathbf{P}_X$. Look for a discrete distribution that assigns positive probability to only three distinct values.

- (a) Let X be a random variable denoting the number of heads that come up in 200 tosses, then $X \sim \text{Bin}(200, \frac{1}{3})$. The event we are interested in is $\{X \geq 120\}$. By Markov's inequality:

$$\mathbf{P}[X \geq 120] \leq \frac{\mathbf{E}[X]}{120} = \frac{200 \cdot \frac{1}{3}}{120} = \frac{5}{9} = 0.\bar{5}.$$

Here we have used that for $Y \sim \text{Bin}(n, p)$, one has

$$(\star) \quad \mathbf{E}[Y] = np.$$

Now we want to apply Čebyšev's inequality. For this, we use that

$$\begin{aligned} \mathbf{P}[X \geq 120] &\leq \mathbf{P}[|X - 200 \cdot \frac{1}{3}| \geq 120 - 200 \cdot \frac{1}{3}] \leq \frac{\text{Var}[X]}{(120 - 200 \cdot \frac{1}{3})^2} = \frac{200 \cdot \frac{1}{3} \cdot \frac{2}{3}}{(120 - 200 \cdot \frac{1}{3})^2} \\ &= \frac{1}{64} = 0.015625. \end{aligned}$$

Here we have used that for $Y \sim \text{Bin}(n, p)$, one has

$$(\star\star) \quad \text{Var}[Y] = np(1-p).$$

Let us also give the calculation for $(\star)$ and $(\star\star)$:¹

$$\begin{aligned} \mathbf{E}[Y] &= \sum_{k=0}^n k \cdot \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} \\ &\stackrel{\ell=k-1}{=} np \underbrace{\sum_{\ell=0}^{n-1} \frac{(n-1)!}{\ell!(n-\ell-1)!} p^\ell (1-p)^{n-1-\ell}}_{=1} = np, \\ \mathbf{E}[Y(Y-1)] &= \sum_{k=0}^n k(k-1) \cdot \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=2}^n \frac{n!}{(k-2)!(n-k)!} p^k (1-p)^{n-k} \\ &\stackrel{\ell=k-2}{=} n(n-1)p^2 \underbrace{\sum_{\ell=0}^{n-2} \frac{(n-2)!}{\ell!(n-\ell-2)!} p^\ell (1-p)^{n-2-\ell}}_{=1} = n(n-1)p^2. \end{aligned}$$

It follows that

$$\text{Var}[Y] = \mathbf{E}[Y^2] - \mathbf{E}[Y]^2 = \mathbf{E}[Y(Y-1)] + \mathbf{E}[Y] - \mathbf{E}[Y]^2 = n^2 p^2 - np^2 - np - n^2 p^2 = np(1-p).$$

In the above computations we implicitly assumed $n \geq 1$ (or $n \geq 2$ for $\mathbf{E}[Y(Y-1)]$). The cases $n = 1$ and $n = 2$ can be treated more directly.

- (b) We consider a random variable $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ attaining the three values $\{-1, 0, 1\}$. The law of X is then completely determined by the three values $p_X(-1)$, $p_X(0)$ and $p_X(1)$, which fulfill $p_X(-1) + p_X(0) + p_X(1) = 1$. Let us assume furthermore that the law is symmetric, i.e. $p_X(-1) = q = p_X(1)$. Then:

$$\begin{aligned} \mathbf{E}[X] &= -1 \cdot q + 0 \cdot (1-2q) + 1 \cdot q = 0, \\ \text{Var}[X] &\stackrel{\mathbf{E}[X]=0}{=} \mathbf{E}[X^2] = (-1)^2 \cdot q + 0^2 \cdot (1-2q) + 1^2 \cdot q = 2q. \end{aligned}$$

We now try to obtain an equality of the form

$$\mathbf{P}[|X - \mathbf{E}[X]| \geq 1] = \frac{\text{Var}[X]}{1^2} \quad \Leftrightarrow \quad \mathbf{P}[X = \pm 1] = 2q = \frac{\text{Var}[X]}{1^2},$$

which is fulfilled for every choice of $q \in (0, \frac{1}{2})$. A concrete example is therefore given by:

$$X : (\{-1, 0, 1\}, \mathcal{P}(\{-1, 0, 1\}), \mathcal{U}(\{-1, 0, 1\})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})), \quad \omega \mapsto \omega,$$

which has the law $\mathbf{P}_X = \mathcal{U}(\{-1, 0, 1\})$, fulfills $\mathbf{E}[X] = 0$, $\text{Var}[X] = \frac{2}{3}$ and

$$\mathbf{P}[|X - \mathbf{E}[X]| \geq 1] = \frac{2}{3} = \frac{\text{Var}[X]}{1^2}.$$

3. $(*)$ Chernoff bounds

[4 Points]

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ a random variable such that the moment generating function $\psi_X(t)$, defined by

$$\psi_X(t) = \mathbf{E}[e^{tX}],$$

exists for every $t \in \mathbb{R}$.

¹Not part of the problem, but sketched for completeness. We will later see an easier way of computing the expectation and variance of a binomial random variable.

- (a) Let $a \in \mathbb{R}$. Show that for $t > 0$, we have

$$\mathbf{P}[X \geq a] \leq e^{-ta} \psi_X(t),$$

whereas for $t < 0$, we have

$$\mathbf{P}[X \leq a] \leq e^{-ta} \psi_X(t).$$

Hint: The function $\exp : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing. Use Markov's inequality.

- (b) Calculate $\psi_X(t)$ for $X \sim \mathcal{N}(0, 1)$.
(c) Show that for $X \sim \mathcal{N}(0, 1)$, we have

$$\mathbf{P}[X \geq a] \leq e^{-\frac{a^2}{2}}, \text{ for } a > 0,$$

and

$$\mathbf{P}[X \leq a] \leq e^{-\frac{a^2}{2}}, \text{ for } a < 0.$$

- (a) We first consider the case where $t > 0$. Note that the function $x \mapsto \exp(tx)$ is strictly increasing, so

$$\mathbf{P}[X \geq a] = \mathbf{P}[e^{tX} \geq e^{ta}] \leq \frac{\mathbf{E}[e^{tX}]}{e^{ta}} = e^{-ta} \psi_X(t).$$

Here we used Markov's inequality for the non-negative random variable e^{tX} . Now for the case of $t < 0$: The function $x \mapsto \exp(-tx)$ is now again strictly increasing, so

$$\mathbf{P}[X \leq a] = \mathbf{P}[-X \geq -a] = \mathbf{P}[e^{tX} \geq e^{ta}] \leq \frac{\mathbf{E}[e^{tX}]}{e^{ta}} = e^{-ta} \psi_X(t).$$

- (b) Let $X \sim \mathcal{N}(0, 1)$:

$$\begin{aligned} \psi_X(t) &= \mathbf{E}[e^{tX}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}x^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}x^2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2} e^{\frac{1}{2}t^2} dx \\ &= e^{\frac{1}{2}t^2} \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz}_{=1} = e^{\frac{1}{2}t^2}. \end{aligned}$$

- (c) Let us again focus on $a > 0$ first. By combining the results of (a) and (b), one has

$$\mathbf{P}[X \geq a] \leq e^{-ta} \psi_X(t) = e^{-ta} e^{\frac{t^2}{2}} = e^{\frac{t^2}{2} - at}.$$

We want the right-hand side to be as small as possible to get the best possible bound. So we optimize the function $u : t \mapsto \frac{t^2}{2} - at$. Clearly, we have $u'(t) = t - a$, $u''(t) = 1$, so we find a minimum at $t = a$ (since $u'(a) = 0$ and $u''(a) > 0$), where

$$\mathbf{P}[X \geq a] \leq e^{\frac{a^2}{2} - a^2} = e^{-\frac{a^2}{2}}.$$

Similarly, for $a < 0$, we can take the result from (a) and find

$$\mathbf{P}[X \leq a] \leq e^{-ta} \psi_X(t) = e^{-ta} e^{\frac{t^2}{2}} = e^{\frac{t^2}{2} - at}.$$

We can then repeat the same calculation as above.

4. (*) Joint distributions

[4 Points]

- (a) For the examples i. and ii. below, decide (with a mathematical explanation) whether the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ can be the joint cumulative distribution function of a pair (X, Y) of random variables. For the case where F is indeed a joint cumulative distribution function, determine the cumulative distribution functions of X and Y .

$$\begin{aligned} \text{i. } F(x, y) &= \begin{cases} 1 - e^{-x-y}, & x, y \geq 0, \\ 0, & \text{otherwise.} \end{cases} \\ \text{ii. } F(x, y) &= \begin{cases} 1 - e^{-x} - xe^{-y}, & 0 \leq x \leq y \\ 1 - e^{-y} - ye^{-x}, & 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

(b) Let X, Y be discrete random variables with joint probability mass function

$$p_{X,Y}(j, k) = \mathbf{P}[X = j, Y = k] = \begin{cases} C \cdot \left(\frac{1}{2}\right)^k, & \text{for } k = 2, 3, \dots, j = 1, 2, \dots, k-1, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the constant C and the marginal probability mass functions p_X and p_Y .

(a) For i., the function is not a joint cumulative distribution function, since

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = -e^{-x-y} < 0, \quad x, y > 0.$$

Therefore, F cannot be a joint cumulative distribution function. For ii., we calculate

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = \begin{cases} e^{-y}, & 0 \leq x \leq y, \\ 0, & 0 \leq y \leq x. \end{cases}$$

We can see that this function satisfies

$$\int_0^\infty \int_0^\infty \frac{\partial^2}{\partial x \partial y} F(x, y) dx dy = 1,$$

so it is a multivariate probability density function. Finally, it is easy to see that

$$\int_{-\infty}^x \int_{-\infty}^y \frac{\partial^2}{\partial x \partial y} F(s, t) dt ds = F(x, y).$$

The cumulative distribution functions of X and Y are

$$\begin{aligned} F_X(x) &= \lim_{y \rightarrow \infty} F(x, y) = 1 - e^{-x}, \quad x \geq 0, \\ F_Y(y) &= \lim_{x \rightarrow \infty} F(x, y) = 1 - e^{-y} - ye^{-y}, \quad y \geq 0. \end{aligned}$$

(b) Since $\mathbf{P}[(X, Y) \in \mathbb{N}^2] = 1$, we see

$$\begin{aligned} 1 &= \sum_{k=2}^{\infty} \sum_{j=1}^{k-1} p_{X,Y}(j, k) = C \sum_{k=2}^{\infty} \sum_{j=1}^{k-1} \left(\frac{1}{2}\right)^k \\ &= C \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^{j+1} \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = C \sum_{j=1}^{\infty} \frac{\left(\frac{1}{2}\right)^{j+1}}{1 - \frac{1}{2}} = C \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^j = C \frac{\frac{1}{2}}{1 - \frac{1}{2}} = C. \end{aligned}$$

We then calculate for $j \geq 1$

$$p_X(j) = \sum_{k=2}^{\infty} p_{X,Y}(j, k) = \sum_{k=j+1}^{\infty} \left(\frac{1}{2}\right)^{j+1} \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^j.$$

On the other hand, for $k \leq 1$ we have $p_X(k) = 0$, and for $k \geq 2$, we have

$$p_Y(k) = \sum_{j=1}^{k-1} p_{X,Y}(j, k) = (k-1) \left(\frac{1}{2}\right)^k.$$